EE6406: Analytic and Ensemble Learning (Lecture 2 Supplement: Mathematics Review)

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Lecture 2
Supplement: Maths Review

Overview

1. Linear Algebra
2. Systems of Linear Equations
3. Functions, Derivative and Gradient
4. Lagrange Optimization

Notations, Vectors and Matrices

- A **scalar** is a simple numerical value, like 68 or -3.13.
- **Variables** or constants that take scalar values are denoted by an italic letter, like x or a.
- We shall focus on real numbers in this course.

Notations, Vectors and Matrices

- The summation over a collection $\{x_1, x_2, x_3, x_4, \cdots, x_m\}$ is denoted like this:

$$\sum_{i=1}^{m} x_i = x_1 + x_2 + \cdots + x_{m-1} + x_m.$$

- The product over a collection $\{x_1, x_2, x_3, x_4, \cdots, x_m\}$ is denoted like this:

$$\prod_{i=1}^{m} x_i = x_1 \cdot x_2 \cdot \cdots \cdot x_{m-1} \cdot x_m.$$

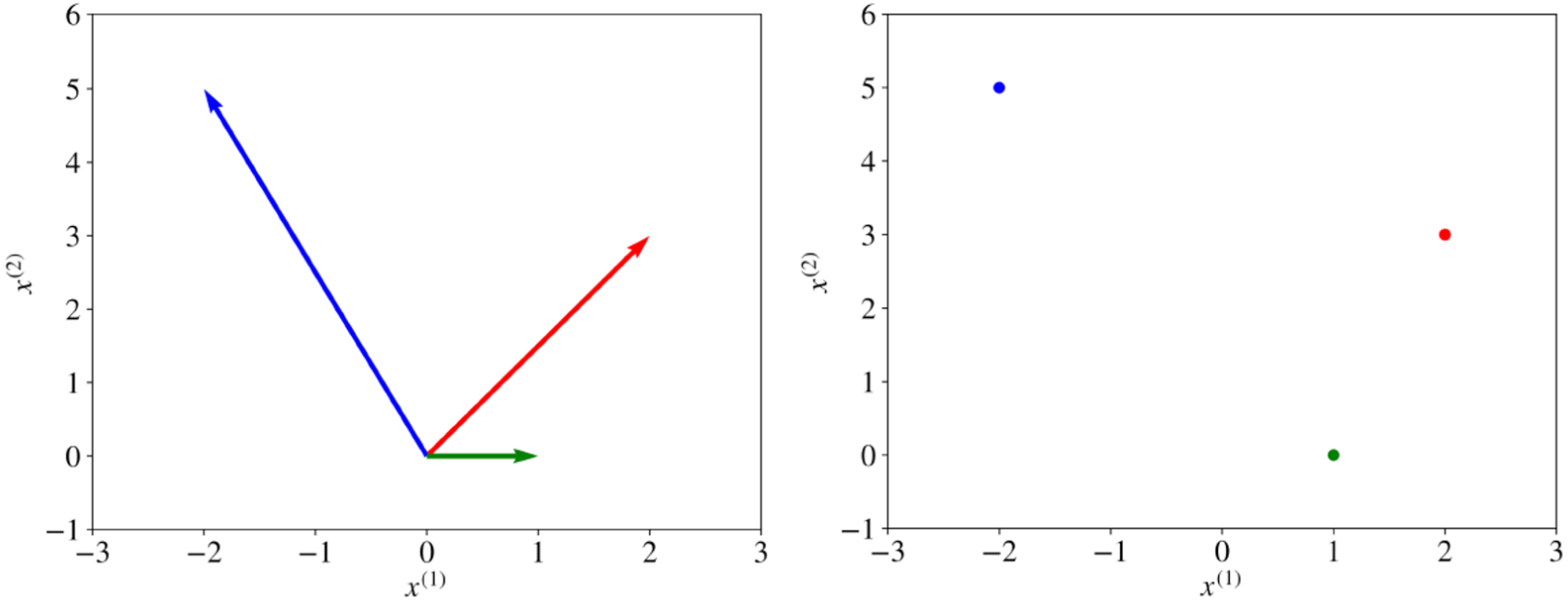
Notations, Vectors and Matrices

- A **vector** is an ordered list of scalar values, called attributes. We denote a vector as a **bold character**, for example, $\mathbf{x}$ or $\mathbf{w}$.
- Vectors can be visualized as arrows that point to some directions as well as points in a multi-dimensional space.
- In many books, vectors are written column-wise:

$$\mathbf{w} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}.$$

Notations, Vectors and Matrices

Illustrations of three two-dimensional vectors, $\mathbf{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$, and $\mathbf{c} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ are given in Figure below.



Three vectors visualized as directions and as points.

Ref: [Book1] Andriy Burkov, “The Hundred-Page Machine Learning Book”, 2019, Chp2.

Notations, Vectors and Matrices

- We denote an attribute of a vector as an italic value with an index, like this: a_j or x_j. The index j denotes a specific dimension of the vector, the position of an attribute in the list. For instance, in the vector **a** shown in red in the Figure of previous page,

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Notations, Vectors and Matrices

- A **matrix** is a rectangular array of numbers arranged in rows and columns. Below is an example of a matrix with two rows and three columns,

$$\mathbf{X} = \begin{bmatrix} 2 & 4 & -3 \\ 21 & -6 & -1 \end{bmatrix}$$

- Matrices are denoted with bold capital letters, such as $\mathbf{A}$, $\mathbf{X}$ or $\mathbf{W}$.

Note:

- A variable can have two or more indices, like this: $x_i^{(j)}$ or like this $x_{i,j}^{(k)}$.
- For example, in neural networks, we denote as $x_{l,u}^{(j)}$ the input feature j of unit u in layer l.
- For the elements in matrix $\mathbf{X}$, we shall use the indexing $x_{i,j}$ where the first and the second indices respectively indicate the row and the column positions.

Notations, Vectors and Matrices

Example

Iris data set

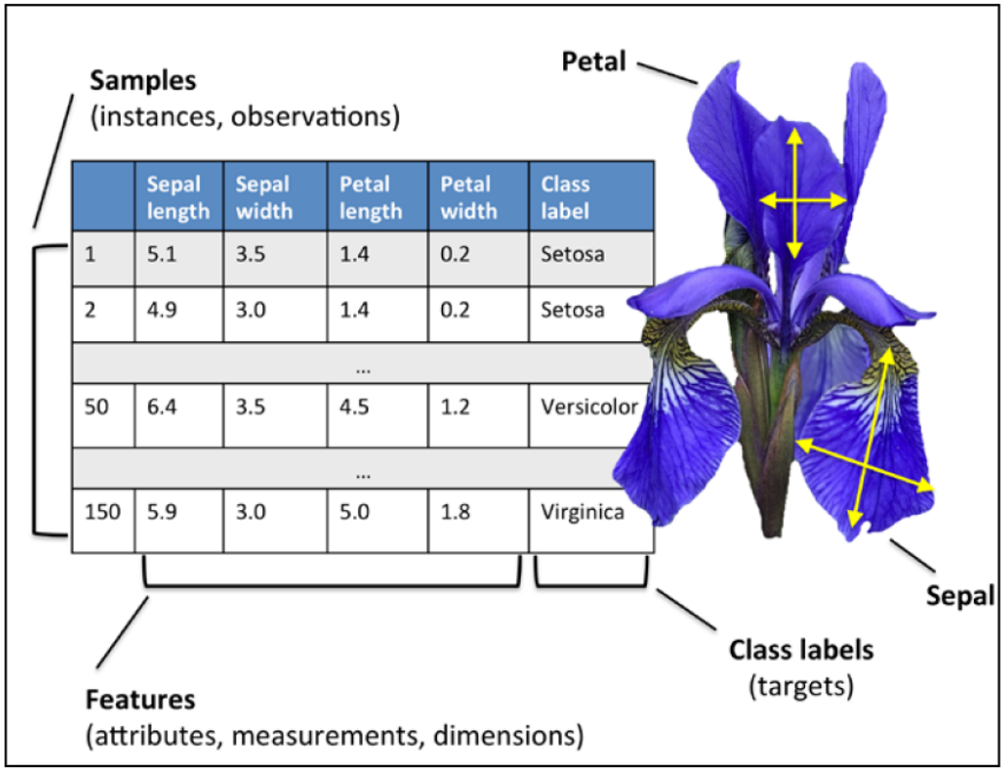
- Measurement features can be packed as

$$\mathbf{X} \in \mathcal{R}^{150\times 4}$$

$$\begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} \\ x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} \\ \vdots & \vdots & \vdots & \vdots \\ x_{150,1} & x_{150,2} & x_{150,3} & x_{150,4} \end{bmatrix}$$

- Labels can be written as

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_{150} \end{bmatrix} \left(y \in \{\text{Setosa, Versicolor, Virginica}\} \right)$$



	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

https://archive.ics.uci.edu/ml/datasets/iris

Notations, Vectors and Matrices

Operations on Vectors:

$$\mathbf{x}+\mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}, \quad \mathbf{x}-\mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \end{bmatrix}$$

$$a\mathbf{x} = a \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 \\ ax_2 \end{bmatrix}, \quad \frac{1}{a}\mathbf{x} = \frac{1}{a} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{a}x_1 \\ \frac{1}{a}x_2 \end{bmatrix}$$

Matrix or Vector Transpose:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{x}^T = [x_1, \ x_2]$$

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \end{bmatrix}, \quad \mathbf{X}^T = \begin{bmatrix} x_{1,1} & x_{2,1} \\ x_{1,2} & x_{2,2} \\ x_{1,3} & x_{2,3} \end{bmatrix}$$

Notations, Vectors and Matrices

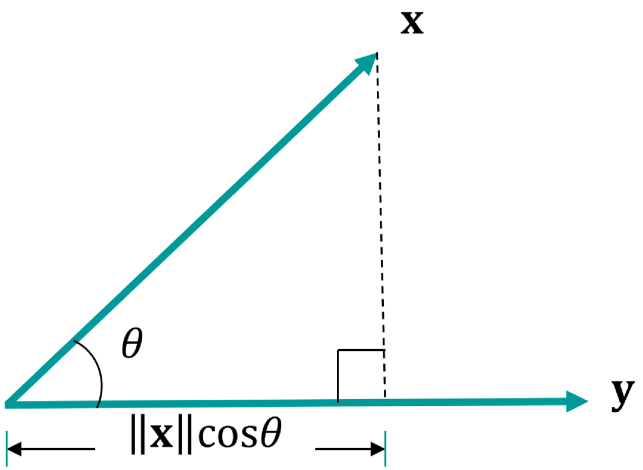
Dot Product or Inner Product of Vectors:

$$\mathbf{x} \cdot \mathbf{y} \quad = \quad \mathbf{x}^T \mathbf{y} = [x_1,\ x_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = x_1 y_1 + x_2 y_2$$

Geometric Definition:

$$\mathbf{y} \cdot \mathbf{x} \quad = \quad \|\mathbf{y}\| \|\mathbf{x}\| \cos(\theta)$$

where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$, and $\|\mathbf{z}\| = \sqrt{\mathbf{z} \cdot \mathbf{z}}$ is the Euclidean length of vector $\mathbf{z}$.



Notations, Vectors and Matrices

Matrix-Vector Product

$$\begin{aligned}
\mathbf{X}\mathbf{w} &= \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \\
&= \begin{bmatrix} x_{1,1}w_1 + x_{1,2}w_2 + x_{1,3}w_3 \\ x_{2,1}w_1 + x_{2,2}w_2 + x_{2,3}w_3 \end{bmatrix}
\end{aligned}$$

Vector-Matrix Product

$$\begin{aligned}
\mathbf{w}^T X &= [w_1 \; w_2 \; w_3] \begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \\ x_{3,1} & x_{3,2} \end{bmatrix} \\
&= [w_1 x_{1,1} + w_2 x_{2,1} + w_3 x_{3,1}, \quad w_1 x_{1,2} + w_2 x_{2,2} + w_3 x_{3,2}]
\end{aligned}$$

Notations, Vectors and Matrices

Matrix-Matrix Product

$$
\begin{aligned}
\mathbf{XW} &= \begin{bmatrix} x_{1,1} & \cdots & x_{1,d} \\ \vdots & \ddots & \vdots \\ x_{m,1} & \cdots & x_{m,d} \end{bmatrix} \begin{bmatrix} w_{1,1} & \cdots & w_{1,h} \\ \vdots & \ddots & \vdots \\ w_{d,1} & \cdots & w_{d,h} \end{bmatrix} \\
&= \begin{bmatrix} (x_{1,1}w_{1,1} + \cdots + x_{1,d}w_{d,1}) & \cdots & (x_{1,1}w_{1,h} + \cdots + x_{1,d}w_{d,h}) \\ \vdots & \ddots & \vdots \\ (x_{m,1}w_{1,1} + \cdots + x_{m,d}w_{d,1}) & \cdots & (x_{m,1}w_{1,h} + \cdots + x_{m,d}w_{d,h}) \end{bmatrix} \\
&= \begin{bmatrix} \sum_{i=1}^{d} x_{1,i}w_{i,1} & \cdots & \sum_{i=1}^{d} x_{1,i}w_{i,h} \\ \vdots & \ddots & \vdots \\ \sum_{i=1}^{d} x_{m,i}w_{i,1} & \cdots & \sum_{i=1}^{d} x_{m,i}w_{i,h} \end{bmatrix}
\end{aligned}
$$

Notations, Vectors and Matrices

Matrix Inverse

Definition:

a d-by-d square matrix $\mathbf{A}$ is said to be invertible (also nonsingular) if there exists a d-by-d square matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$ (identity matrix) given by

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

of d-by-d dimension.

Notations, Vectors and Matrices

Matrix inverse computation

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A})$$

- $\det(\mathbf{A})$ is the determinant of $\mathbf{A}$.
- $\operatorname{adj}(\mathbf{A})$ is the adjugate or adjoint of $\mathbf{A}$ which is the transpose of its cofactor matrix.
- The (i, j) entry of the cofactor matrix $\mathbf{C}$ is the (i, j)-minor times a sign factor. E.g., $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\operatorname{adj}(\mathbf{A}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \mathbf{C}^T$.

Notations, Vectors and Matrices

Determinant computation

Example: 2×2 matrix

$$\det(\mathbf{A}) = |\mathbf{A}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example: 3×3 matrix

$$|A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} \square & \square & \square \\ \square & e & f \\ \square & h & i \end{vmatrix} - b \begin{vmatrix} \square & \square & \square \\ d & \square & f \\ g & \square & i \end{vmatrix} + c \begin{vmatrix} \square & \square & \square \\ d & e & \square \\ g & h & \square \end{vmatrix}$$

$$= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a(ei - fh) - b(di - fg) + c(dh - eg)$$

Notations, Vectors and Matrices

Cofactor matrix computation

Consider a 3×3 matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

The minor $\mathbf{A}_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

Its cofactor matrix is

$$\mathbf{C} = \begin{pmatrix} +\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & +\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & +\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ +\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & +\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{pmatrix}.$$

Notations, Vectors and Matrices

Example

Find the cofactor matrix of $\mathbf{A}$ given that $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$.

Solution:

$$a_{11} = \begin{vmatrix} 4 & 5 \\ 0 & 6 \end{vmatrix} = 24, \quad a_{12} = -\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} = 5, \quad a_{13} = \begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix} = -4,$$

$$a_{21} = -\begin{vmatrix} 2 & 3 \\ 0 & 6 \end{vmatrix} = -12, \quad a_{22} = \begin{vmatrix} 1 & 3 \\ 1 & 6 \end{vmatrix} = 3, \quad a_{23} = -\begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = 2,$$

$$a_{31} = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -2, \quad a_{32} = -\begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix} = -5, \quad a_{33} = \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 4,$$

The cofactor matrix is thus $\mathbf{C} = \begin{bmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{bmatrix}$.

Linear Dependency

Linear Dependence

- A collection of d-vectors $\mathbf{x}_1, \cdots, \mathbf{x}_m$ (with $m \geqslant 1$) is called linearly dependent if

$$\beta_1 \mathbf{x}_1 + \cdots + \beta_m \mathbf{x}_m = 0$$

holds for some $\beta_1, \cdots, \beta_m$ that are not all zero.

Note: a vector of size/dimension/length d is called a d-vector. $\beta_1, \cdots, \beta_m$ are scalars.

Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (Chp5)

Linear Dependency

Linear Independence

- A collection of d-vectors $\mathbf{x}_1, \cdots, \mathbf{x}_m$ (with $m \geqslant 1$) is called linearly independent if it is not linearly dependent, which means that

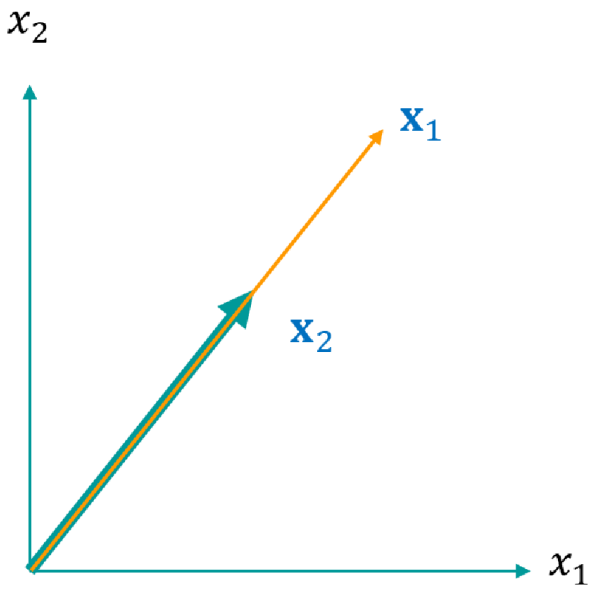
$$\beta_1 \mathbf{x}_1 + \cdots + \beta_m \mathbf{x}_m = 0$$

only holds for $\beta_1 = \cdots = \beta_m = 0$.

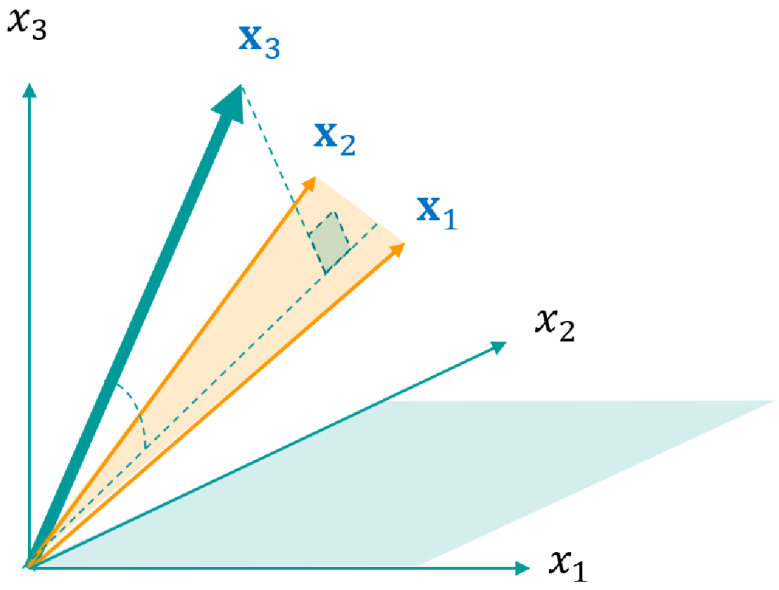
Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (Chp5)

Linear Dependency

Geometry of dependency and independency



$$\beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 = 0$$



$$\beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 \neq \beta_3\mathbf{x}_3$$

Set and Function

- A set is an unordered collection of unique elements.
- We denote a set as a calligraphic capital character, for examples, $\mathbb{R}, \mathbb{N}, \mathbb{C}, \mathcal{R}, \mathcal{N}, \mathcal{C}$ etc.
- A set of numbers can be finite (include a fixed amount of values). In this case, it is denoted using accolades, for example, $\{1, 3, 18, 23, 235\}$ or $\{x_1, x_2, x_3, x_4, \cdots, x_d\}$.
- A set can also be infinite.

Set and Function

- A set can be infinite and include all values in some interval.
- If a set includes all values between a and b, including a and b, it is denoted using brackets as $[a,\ b]$.
- If the set does not include the values a and b, such a set is denoted using parentheses like this: $(a,\ b)$.
- For example, the set $[0,\ 1]$ includes such values as 0, 0.0001, 0.25, 0.784, 0.9995, and 1.0.
- A special set denoted $\mathbb{R}$ (or $\mathcal{R}$) includes all real numbers from minus infinity to plus infinity.

Set and Function

- Intersection of two sets:

$$\mathcal{S}_3 \leftarrow \mathcal{S}_1 \cap \mathcal{S}_2$$

Example: $\{1, 3, 5, 8\} \cap \{1, 8, 4\} = \{1, 8\}$

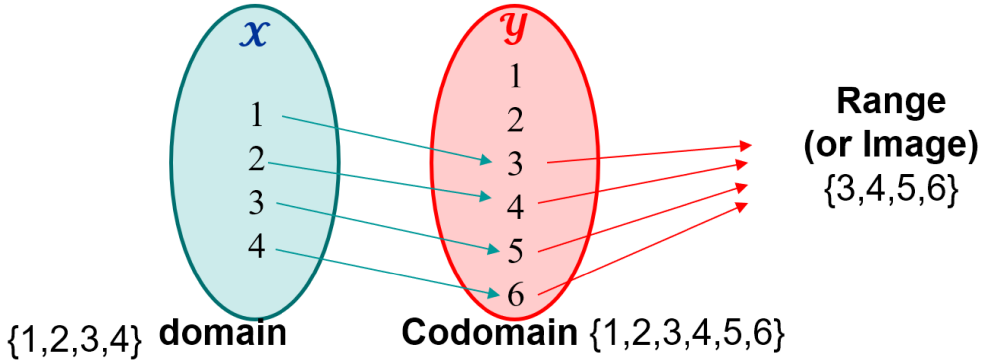
- Union of two sets:

$$\mathcal{S}_3 \leftarrow \mathcal{S}_1 \cup \mathcal{S}_2$$

Example: $\{1, 3, 5, 8\} \cup \{1, 8, 4\} = \{1, 3, 4, 5, 8\}$

Set and Function

- A function is a relation that associates each element x of a set $\mathcal{X}$, the domain of the function, to a single element y of another set $\mathcal{Y}$, the codomain of the function.
- A function usually has a name. If the function is called f, this relation is denoted $y = f(x)$ (read f of x), the element x is the argument or input of the function, and y is the value of the function or the output.
- The symbol that is used for representing the input is the variable of the function (we often say that f is a function of the variable x).



Set and Function

- A scalar function can also have vector argument such as, $y = f(\mathbf{x})$.
- A vector function, denoted as $\mathbf{y} = \mathbf{f}(\mathbf{x})$ is a function that returns a vector $\mathbf{y}$.
- It can have a vector ($\mathbf{y} = \mathbf{f}(\mathbf{x})$) or a scalar argument ($\mathbf{y} = \mathbf{f}(x)$).
- The notation $f : \mathbb{R}^d \to \mathbb{R}$ means that f is a function that maps real d-vectors to real numbers, i.e., it is a scalar-valued function of d-vectors. If $\mathbf{x}$ is a d-vector, then $f(\mathbf{x})$, which is a scalar, denotes the value of the function f at $\mathbf{x}$. (In the notation $f(\mathbf{x})$, $\mathbf{x}$ is referred to as the argument of the function.)

$$f(\mathbf{x}) = f(x_1, x_2, \cdots, x_d)$$

Set and Function

Linear and Affine Functions

- To describe a function $f : \mathbb{R}^d \to \mathbb{R}$, we have to specify what its value is for any possible argument $\mathbf{x} \in \mathbb{R}^d$.
- For example, we can define a function $f : \mathbb{R}^4 \to \mathbb{R}$ by

$$f(\mathbf{x}) = x_1 + x_2 - x_4^2.$$

Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (p32)

Set and Function

Linear and Affine Functions

Another example: The inner product function

- Suppose $\mathbf{w}$ is a d-vector. We can define a scalar valued function f of d-vectors, given by

$$f(\mathbf{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d. \tag{1}$$

  for any d-vector $\mathbf{x}$.

- This function gives the inner product of its d-vector argument $\mathbf{x}$ with some (fixed) d-vector $\mathbf{w}$.
- We can also think of f as forming a weighted sum of the elements of $\mathbf{x}$; the elements of $\mathbf{w}$ give the weights used in forming the weighted sum.

Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (p32)

Set and Function

Linear and Affine Functions

A function $f : \mathbb{R}^d \to \mathbb{R}$ is **linear** if it satisfies the following two properties.

- **Homogeneity**: For any d-vector $\mathbf{x}$ and any scalar α, $f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$.
- **Additivity**: For any d-vectors $\mathbf{x}$ and $\mathbf{y}$, $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$.

Homogeneity states that scaling the (vector) argument is the same as scaling the function value.
Additivity says that adding (vector) arguments is the same as adding the function values.

Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (p32)

Set and Function

Linear and Affine Functions

Superposition and linearity.

- The inner product function f defined in (1) (see slide 45) satisfies the property

$$\begin{aligned} f(\alpha\mathbf{x}+\beta\mathbf{y}) &= \mathbf{w}^T(\alpha\mathbf{x}+\beta\mathbf{y}) \\ &= \mathbf{w}^T(\alpha\mathbf{x})+\mathbf{w}^T(\beta\mathbf{y}) \\ &= \alpha(\mathbf{w}^T\mathbf{x})+\beta(\mathbf{w}^T\mathbf{y}) \\ &= \alpha f(\mathbf{x})+\beta f(\mathbf{y}) \end{aligned}$$

  for all d-vectors $\mathbf{x}$, $\mathbf{y}$, and all scalars α, β.

- This property is called superposition (which consists of homogeneity and additivity).
- A function that satisfies the superposition property is called linear.

Set and Function

Linear and Affine Functions

If a function f is linear, superposition extends to linear combinations of any number of vectors:

$$f(\alpha_1 \mathbf{x}_1 + \cdots + \alpha_k \mathbf{x}_k) = \alpha_1 f(\mathbf{x}_1) + \cdots + \alpha_k f(\mathbf{x}_k)$$

for any d vectors $\mathbf{x}_1, \cdots, \mathbf{x}_k$, and any scalars $\alpha_1, \cdots, \alpha_k$.

Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (p32)

Set and Function

Linear and Affine Functions

A linear function $f : \mathbb{R}^d \to \mathbb{R}$ is affine if and only if it can be expressed as $f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b$ for some d-vector $\mathbf{w}$ and scalar b, which is sometimes called the offset.

Example

$$f(\mathbf{x}) = 2.3 - 2x_1 + 1.3x_2 - x_3$$

is affine, with $b = 2.3$, $\mathbf{w}^T = [-2,\ 1.3,\ -1]$.

Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (p32)

Set and Function

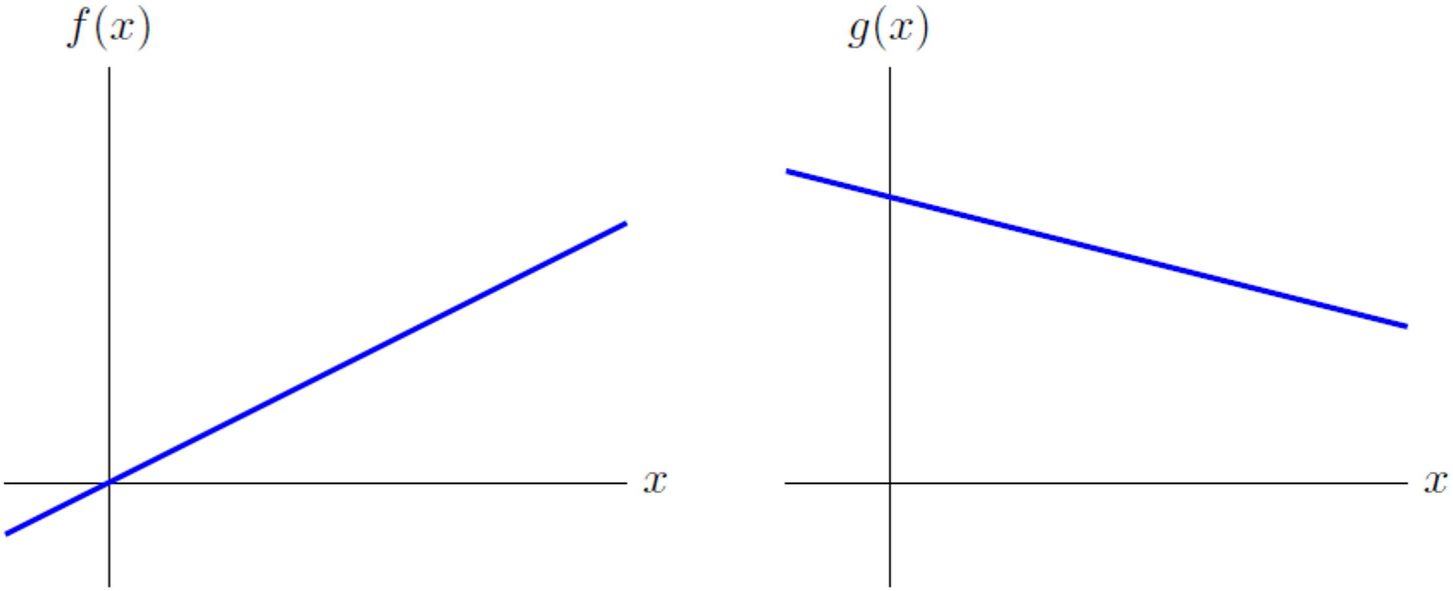


Figure 2.1 *Left.* The function f is linear. *Right.* The function g is affine, but not linear.

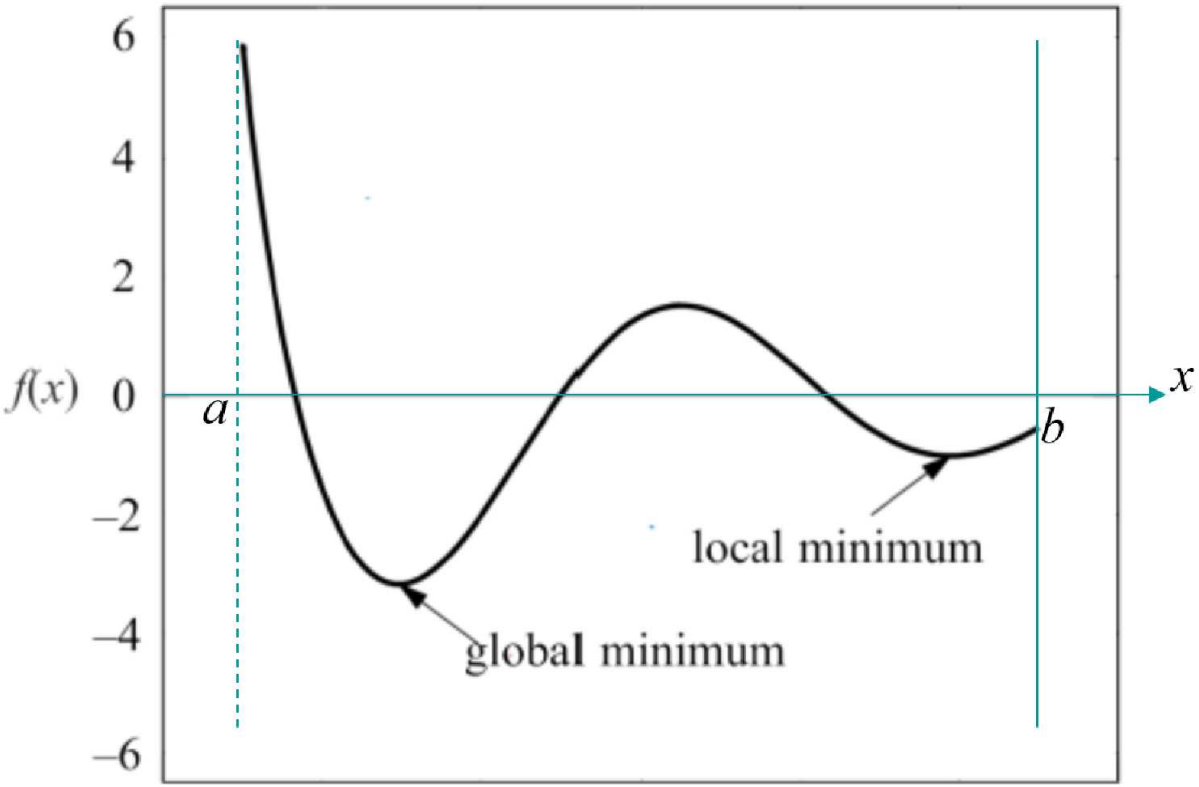
Ref: Stephen Boyd and Lieven Vandenberghe, "Introduction to Applied Linear Algebra", Cambridge University Press, 2018 (p32)

Set and Function

- We say that $f(x)$ has a **local minimum** at $x = c$ if $f(x) \geqslant f(c)$ for every x in some open interval around $x = c$.
- An interval is a set of real numbers with the property that any number that lies between two numbers in the set is also included in the set.
- An open interval does not include its endpoints and is denoted using parentheses. For example, (0, 1) means "all numbers greater than 0 and less than 1".
- The minimal value among all the local minima is called the global minimum. See illustration in the figure on next page.

Set and Function

Global and Local Minima



A local and a global minima of a function.

$a < x \leq b$

Ref: [Book1] Andriy Burkov, "The Hundred-Page Machine Learning Book", 2019 (p6-7 of chp2).

Set and Function

- Given a set of values $\mathcal{A} = \{a_1, a_2, \cdots, a_m\}$, the operator $\max_{a \in \mathcal{A}} f(a)$ returns the highest value $f(a)$ for all elements in the set $\mathcal{A}$.
- On the other hand, the operator $\arg\max_{a \in \mathcal{A}} f(a)$ returns the element of the set $\mathcal{A}$ that maximizes $f(a)$.
- Sometimes, when the set is implicit or infinite, we can write

$$\max_a f(a) \quad \text{or} \quad \arg\max_a f(a)$$

- Operators "min" and "arg min" operate in a similar manner.

Note: **arg max** returns a value from the domain of the function and **max** returns from the range (codomain) of the function.

Derivative and Gradient

- A derivative f' of a function f is a function or a value that describes how fast f grows (or decreases).
- If the derivative is a constant value, like 5 or -3, then the function grows (or decreases) constantly at any point x of its domain.
- If the derivative f' is positive at some point x, then the function f grows at this point.
- If the derivative of f is negative at some x, then the function decreases at this point.
- The derivative of zero at x means that the function's slope at x is horizontal.

Derivative and Gradient

Partial Derivatives

Differentiation of a scalar function w.r.t. a vector.

If $f(\mathbf{x})$ is a scalar function of d variables and $\mathbf{x}$ is a $d \times 1$ vector. Then differentiation of $f(\mathbf{x})$ with respect to $\mathbf{x}$ results in a $d \times 1$ vector:

$$\frac{df(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

This is referred to as the **gradient** of $f(\mathbf{x})$ and often written as $\nabla_{\mathbf{x}} f$.

Derivative and Gradient

Partial Derivatives

Differentiation of a vector function w.r.t. a vector.

If $\mathbf{f}(\mathbf{x})$ is a vector function of size $h \times 1$ and $\mathbf{x}$ is a $d \times 1$ vector. Then differentiation of $\mathbf{f}(\mathbf{x})$ with respect to $\mathbf{x}$ results in a $h \times d$ matrix:

$$\frac{d\mathbf{f}(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_h}{\partial x_1} & \cdots & \frac{\partial f_h}{\partial x_d} \end{bmatrix}$$

The matrix is referred to as the **Jacobian** of $\mathbf{f}(\mathbf{x})$.

Derivative and Gradient

Some Vector-Matrix Differentiation Formulae

$$\begin{aligned}
\frac{d\mathbf{A}\mathbf{x}}{d\mathbf{x}} &= \mathbf{A} \\
\frac{d\mathbf{y}^T\mathbf{A}\mathbf{x}}{d\mathbf{x}} &= \mathbf{A}^T\mathbf{y} \\
\frac{d\mathbf{x}^T\mathbf{A}\mathbf{x}}{d\mathbf{x}} &= (\mathbf{A}+\mathbf{A}^T)\mathbf{x}
\end{aligned}$$

Constrained Optimization

Suppose we seek the minimum position ($\mathbf{x}_0$) of a scalar-valued function $f(\mathbf{x})$, subject to some constraint. If a constraint can be expressed in the form $g(\mathbf{x}) = 0$, then we can find the extremum of $f(\mathbf{x})$ as follows.
First we form the Lagrangian function

$$L(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda g(\mathbf{x}), \tag{2}$$

where λ is a scalar called the *Lagrange undetermined multiplier*.
We convert this constrained optimization problem into an unconstrained problem by taking the derivative,

$$\frac{\partial L(\mathbf{x}, \lambda)}{\partial \mathbf{x}} = \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} + \lambda \frac{\partial g(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{0} \tag{3}$$

and using standard methods from calculus to solve the resulting equations for λ and the extremizing value of $\mathbf{x}$.

Thank you!